Polynomials — problem 10.68

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Let $P_1(x) = x^2 - 2$ and $P_j(x) = P_1\left(P_{j-1}(x)\right)$ for $j = 2, 3, \ldots$. Show that, for any positive integer n, the roots of the equation $P_n(x) = x$ are real and distinct. (IMO 1976.)